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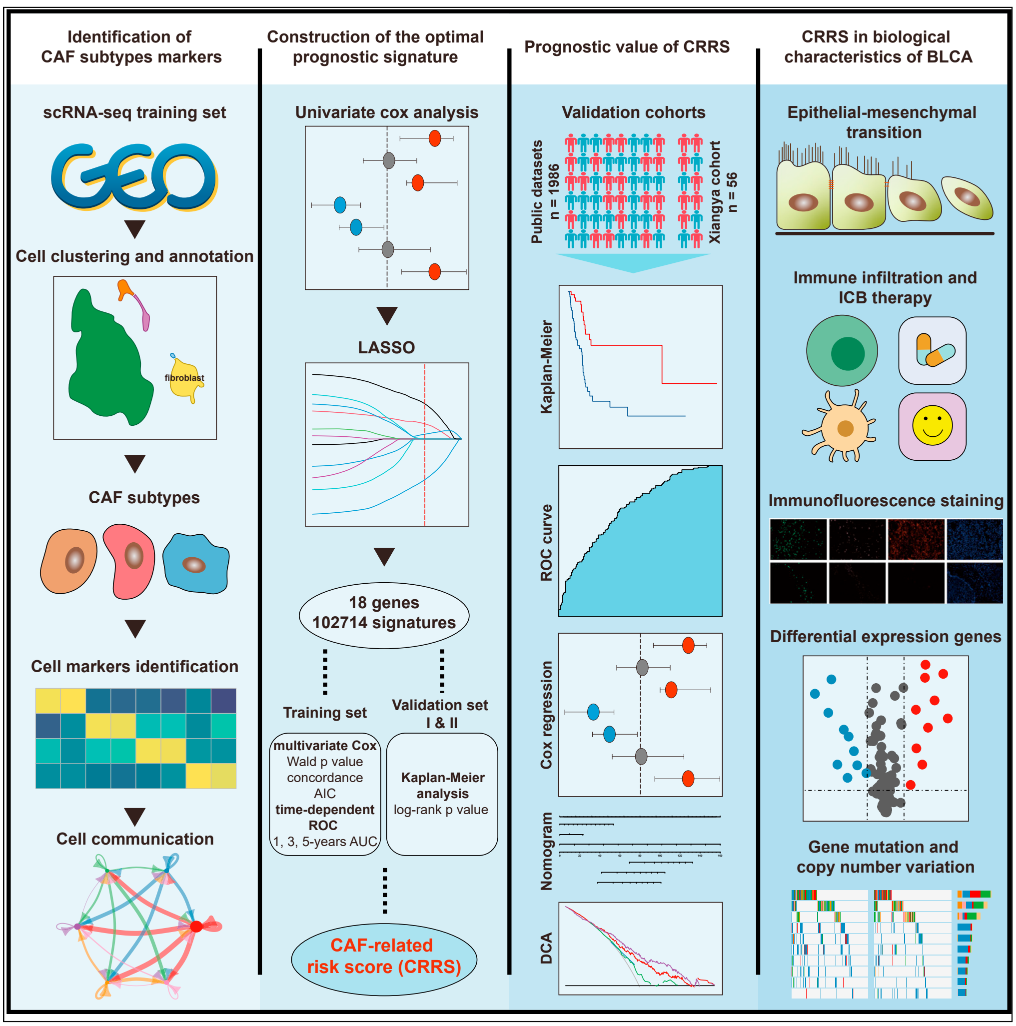

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Highlights

Identification of CAF subtypes-related markers according to single-cell RNA-seq data

Construction of CAFs-related risk score (CRRS) by screening 102,714 signatures

CRRS was a valuable predictor in multiple validation cohorts

CRRS was significantly correlated with biological characteristics of BLCA

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A cancer-associated fibroblast subtypes-based signature enables the evaluation of immunotherapy response and prognosis in bladder cancer

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SUMMARY

Bladder cancer (BLCA) is one of the most prevalent and heterogeneous urinary malignant tumors. Previous researches have reported a significant association between cancer-associated fibroblasts (CAFs) and poor prognosis of tumor patients. However, uncertainty surrounds the role of CAFs in the BLCA tumor microenvironment, necessitating further investigation into the CAFs-related gene signatures in BLCA. In this study, we identified three CAF subtypes in BLCA according to single-cell RNA-seq data and constructed CAFs-related risk score (CRRS) by screening 102,714 signatures. The survival analysis, ROC curves, and nomogram suggested that CRRS was a valuable predictor in 2,042 patients from 9 available public datasets and Xiangya real-world cohort. We further revealed the significant correlation between CRRS and clinicopathological characteristics, genome alterations, and epithelial-mesenchymal transition (EMT). A high CRRS indicated a non-inflamed phenotype and a lower remission rate of immunotherapy in BLCA. In conclusion, the CRRS had the potential to predict the prognosis and immunotherapy response of BLCA patients.

INTRODUCTION

Bladder cancer (BLCA) is one of the most common urinary malignant tumors and is becoming an increasingly serious threat to human health.[1] BLCA is a highly heterogeneous disease which is characterized by rapid progress and a high recurrence rate.[2] Despite advancements in chemotherapy and targeted therapy, metastatic BLCA still has a poor prognosis.[3] Therefore, developing new effective tools to predict prognosis and therapeutic response are particularly significant for patients suffering from BLCA.

The tumor microenvironment (TME) has drawn extensive attention in recent years. As an important component of TME, cancer-associated fibroblasts (CAFs) can secrete a variety of cytokines and extracellular matrix (ECM) proteins to promote immune evasion, therapeutic resistance and the development of cancers.[4] Previous research has reported a significant association between high levels of CAFs and a poor prognosis in tumor patients.[5,6] However, the role of CAFs in the BLCA tumor microenvironment is complex, and the CAF-related gene signatures in BLCA are required to deeply explore the role of CAFs in the BLCA tumor microenvironment.

The single-cell RNA sequencing (scRNA-seq) is a forceful technology used to disclose tumor heterogeneity and TME status.[7] In the last few years, scRNA-seq has been used to classify CAFs into different subtypes and investigate the distribution of CAFs in cancers.[8,9] Zhaohui et al. first reported that inflammatory cancer-associated fibroblasts (iCAFs) were an important factor in BLCA progression by comprehensively profiling the 52721 single cells from primary bladder tumor tissues or paratumor mucosae.[10] More researches are needed to validate the influence of other CAF subtypes on BLCA development.

In the present study, using scRNA-seq data, we performed a comprehensive analysis of the CAFs subtypes in BLCA patients. Subsequently, we constructed a CAF-correlated prognostic signature for predicting the overall survival (OS) of BLCA patients. The accuracy of our CAF-related risk score (CRRS) was evaluated using multiple datasets. We further evaluated the connection between CRRS and clinicopathological characteristics, epithelial-mesenchymal transition (EMT), immune phenotypes and therapeutic responsiveness of BLCA. To summarize, our study suggested that CRRS had the potential to predict the outcomes as well as the immunotherapy response of BLCA patients.

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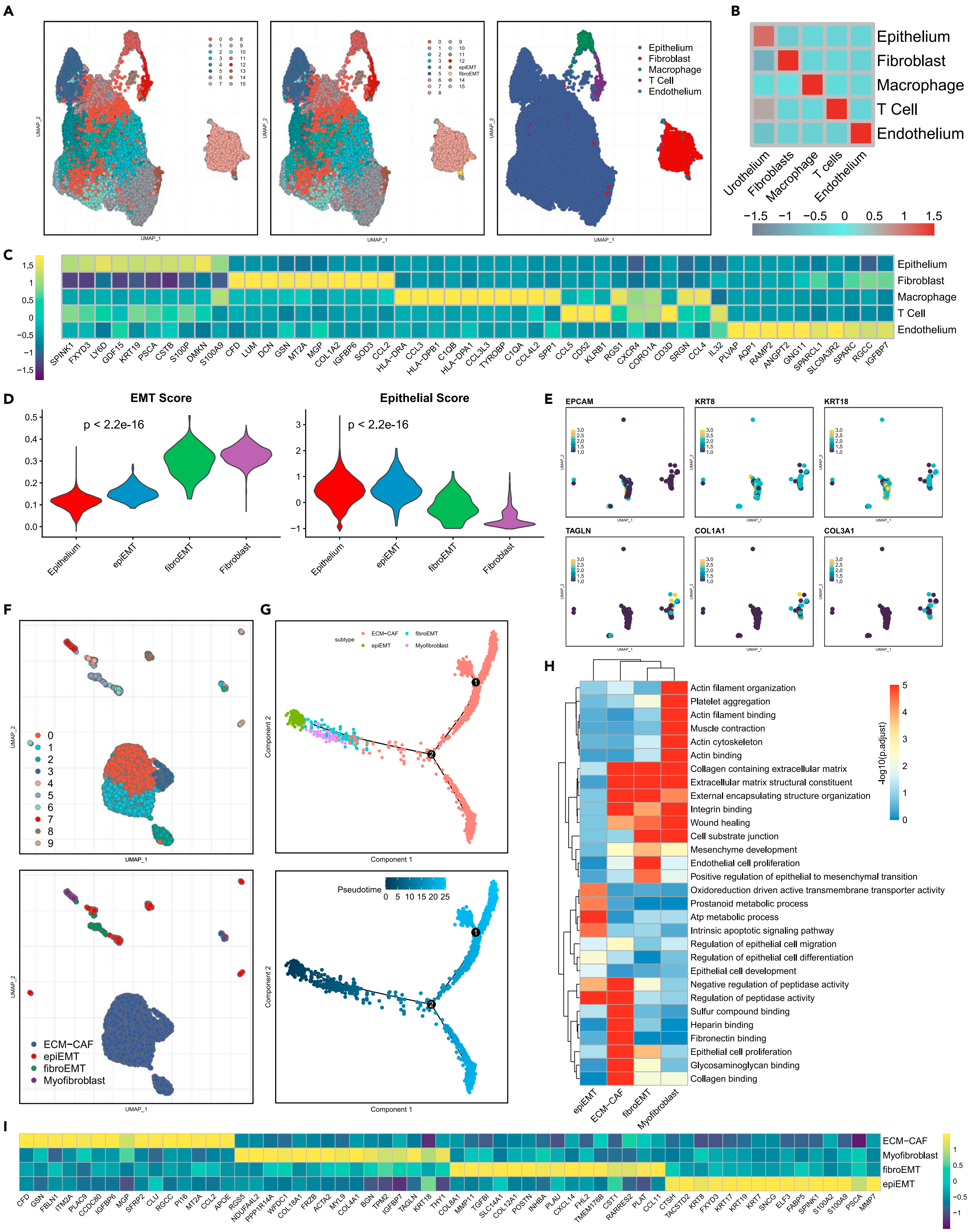
A
UMAP_1
UMAP_2
Epithelium
Fibroblast
Macrophage
T Cell
Endothelium
epiEMT
fibroEMT
B
Epithelium
Fibroblast
Macrophage
T Cell
Endothelium
Urothelium
Fibroblasts
Macrophage
T cells
Endothelium
−1.5 −1 −0.5 0 0.5 1 1.5
C
SPINK1 FXYD3 LY6D GDF15 KRT19 PSCA CSTB S100P DMKN S100A9 CFD LUM DCN GSN MT2A MGP COL1A2 IGFBP6 SOD3 CCL2 HLA−DRA CCL3 HLA−DPB1 C1QB HLA−DPA1 CCL3L3 TYROBP C1QA CCL4L2 SPP1 CCL5 CD52 KLRB1 RGS1 CXCR4 CORO1A CD3D SRGN CCL4 IL32 PLVAP AQP1 RAMP2 ANGPT2 GNG11 SPARCL1 SLC9A3R2 SPARC RGCC IGFBP7
D
EMT Score
p < 2.2e-16
Epithelial Score
p < 2.2e-16
Epithelium
epiEMT
fibroEMT
Fibroblast
E
EPCAM
KRT8
KRT18
TAGLN
COL1A1
COL3A1
F
ECM−CAF
epiEMT
fibroEMT
Myofibroblast
G
subtype
ECM−CAF
fibroEMT
epiEMT
Myofibroblast
Component 1
Component 2
Pseudotime
0 5 10 15 20 25
H
-log10(p.adjust)
Actin filament organization
Platelet aggregation
Actin filament binding
Muscle contraction
Actin cytoskeleton
Actin binding
Collagen containing extracellular matrix
Extracellular matrix structural constituent
External encapsulating structure organization
Integrin binding
Wound healing
Cell substrate junction
Mesenchyme development
Endothelial cell proliferation
Positive regulation of epithelial to mesenchymal transition
Oxidoreduction driven active transmembrane transporter activity
Prostanoid metabolic process
Atp metabolic process
Intrinsic apoptotic signaling pathway
Regulation of epithelial cell migration
Regulation of epithelial cell differentiation
Epithelial cell development
Negative regulation of peptidase activity
Regulation of peptidase activity
Sulfur compound binding
Heparin binding
Fibronectin binding
Epithelial cell proliferation
Glycosaminoglycan binding
Collagen binding
epiEMT
ECM−CAF
fibroEMT
Myofibroblast
I
ECM−CAF
Myofibroblast
fibroEMT
epiEMT
CFD GSN FBLN1 ITM2A PLAC9 CCDC80 IGFBP6 MGP SFRP2 CLU RGCC PI16 MT2A CCL2 APOE RGS5 NDUFA4L2 PPP1R14A WFDC1 COL18A1 FRZB ACTA2 MYL9 COL4A1 BGN TPM2 IGFBP7 TAGLN KRT18 THY1 COL8A1 MMP11 TGFBI SLC14A1 COL12A1 POSTN INHBA PLAU CXCL14 FHL2 TMEM176B CST1 RARRES2 PLAT CCL11 CTSH TACSTD2 KRT8 FXYD3 KRT17 KRT19 KRT7 SNCG ELF3 FABP5 SPINK1 S100A2 S100A9 PSCA MMP7

Figure 1. Single-cell transcriptome analysis and the cell types identification in human bladder cancer

(A) UMAP plotting of the 42440 cells showing 16 cell clusters and 5 distinct cell types identified by marker genes; (B) The corresponding cell type score in each annotated cell type.
(C) Heatmap for the top 10 DEGs in each cell type.
(D) The EMT score and the epithelial score of epithelium, epiEMT, fibroEMT and fibroblast.
(E) The expression of epithelial markers and the stromal markers in epiEMT (up) and fibroEMT (down) cell clusters.
(F) UMAP plot of fibroblast cells.
(G) Trajectory analysis of epiEMT and CAF subtypes along pseudotime by Monocle2.
(H) GSEA analysis of GO pathways in epiEMT and CAF subclusters.
(I) Heatmap for the top 15 DEGs in epiEMT and CAF subclusters.
See also Figures S1 and S2; Tables S2, S3, and S6.

RESULTS

Single-cell transcriptomic profiling and CAF subtypes identification in human bladder cancer

Seven tumor tissues and one paratumor tissue from seven muscle-invasive BLCA patients were collected to explore the cellular heterogeneity and relevant molecular characteristics.[11] After quality control and removal of batch effect, 42440 single cells were obtained for further analysis. After dimensional reduction and cell clustering, 16 clusters were obtained based on the optimal number of principal components (PC = 20) and resolution (R = 0.5; see "STAR Methods"), and the cells of cluster 13 were further divided into epiEMT and fibroEMT (Figure 1A). The clusters were assigned to five major cell types based on common markers for epithelial cells (marked by EPCAM, KRT8, KRT18), macrophage (marked by CD14, CSF1R, AIF1), T cells (marked by CD2, CD3D, CD3E), fibroblasts (marked by DCN, PDPN, TAGLN) and endothelial cells (marked by PECAM1, VWF, CLDN5) (Figures 1A and S1A–S1E). The cell type scores of each annotated cell type were calculated using corresponding common markers (Figure 1B). Afterward, we conducted differentially expressed gene analysis to identify the characteristics of the five cell types, and the top 10 genes significantly differentially expressed (SDE) for each cell type were represented in the heatmap as shown in Figure 1C. EMT is the process whereby epithelial cells transform into mesenchymal cells and plays an important role in tumor invasion and metastasis. The EMT score and epithelial score of the epithelium, epiEMT, fibroEMT and fibroblast were conducted using the "addmodulescore" function of Seurat package. The results indicated that EMT score increased gradually from epithelial and epiEMT to fibroEMT and fibroblast, while the epithelial score decreased (Figure 1D). The epithelial markers EPCAM, KRT8, and KRT18 were obviously enriched in the epiEMT, whereas the stromal markers including TAGLN, COL1A1, and COL3A1 were highly expressed in the fibroEMT (Figure 1E).

CAFs had been confirmed as a significant component of the tumor microenvironment and was considered to be crucial in facilitating cancer progression.[12] In this study, 2362 fibroblast cells were classified into three distinct subclusters, including ECM-CAF, fibroEMT and myofibroblast based on the expression of specific cellular markers (Figure 1F; Table S2). Furthermore, we performed cell trajectory analysis to confirm the developmental states of CAFs. Along the trajectory of pseudotime from early to late, the number of epiEMT cells decreased, while the number of ECM-CAF cells increased (Figure 1G). To represent the characteristics of CAF subtypes, GSEA was performed for each CAF subtype and the representative pathways selected were shown in Figure 1H and Table S3. As expected, pathways that were related to the extracellular matrix were enriched in all three CAF subtypes; pathways that were related to actin and muscle contraction were enriched in myofibroblast; whereas pathways related to cell substrate and EMT were significantly enriched in fibroEMT. Besides, the heatmap of the top 15 differentially expressed genes of each CAF subtypes was presented in Figure 1I.

Cell communication analyses were conducted to investigate the frequent communication between different CAF subtypes and immune cells (Figure S2A). We found that CAFs subtypes were strongly associated with T cells (Figure S2B). In addition, CAFs may influence the tumor microenvironment owing to greater interactions with immune cells through ligand-receptor pairs such as APP-CD74, COL1A2-SDC1, etc., which indicated that blocking the communication of CAFs with immune cells might be an effective therapeutic measure for BLCA (Figure S2C).

Construction of CAFs-related risk score in BLCA

Univariate Cox regression analysis was performed to screen CAF-associated prognostic genes in the TCGA training cohort (Table S4A). We then confirmed 18 markers with minimal lambda (0.03243242) using the LASSO algorithm (Figure 2A; Table S4B). To further acquire the optimal prediction model, we constructed 102714 signatures based on the 18 candidate markers. Then, we conducted multivariate Cox regression and time-dependent ROC curve on TCGA training cohort and tested log rank p value in two external validation cohorts (IMvigor210 and GSE13507) for each signature (Table S4C). Finally, 8 CAF-related markers were selected as the optimal prognostic signature for patients with BLCA comprehensively (Figure 2B). We further validated the expression of the 8 markers selected on the single-cell training set GSE135337 and scRNA-seq validation dataset GSE130001. The results indicated that EMP1 was mainly expressed in the ECM-CAF subpopulation, MEST and P4HB were mainly expressed in the fibroEMT subpopulation, and other markers including CDH6, COX4I2, EGFL6, CRIP2 and MAP1B were mainly expressed in the myofibroblast subpopulation (Figures S3A–S3E). CAF-related risk score (CRRS) was calculated based on the expression level and multivariate Cox regression coefficient of the 8 markers. As shown in Figure 2C, patients with high-CRRS had a significantly worse prognosis than those with low-CRRS. Furthermore, we successfully validated the effect of the CRRS on predicting the overall survival time of the TCGA training and validation cohorts (Figure 2D). Compared with other clinicopathological parameters, the ROC curves indicated that the CRRS had a stronger predictive capacity (Figure 2E). The AUCs of ROC curves in CRRS predicting

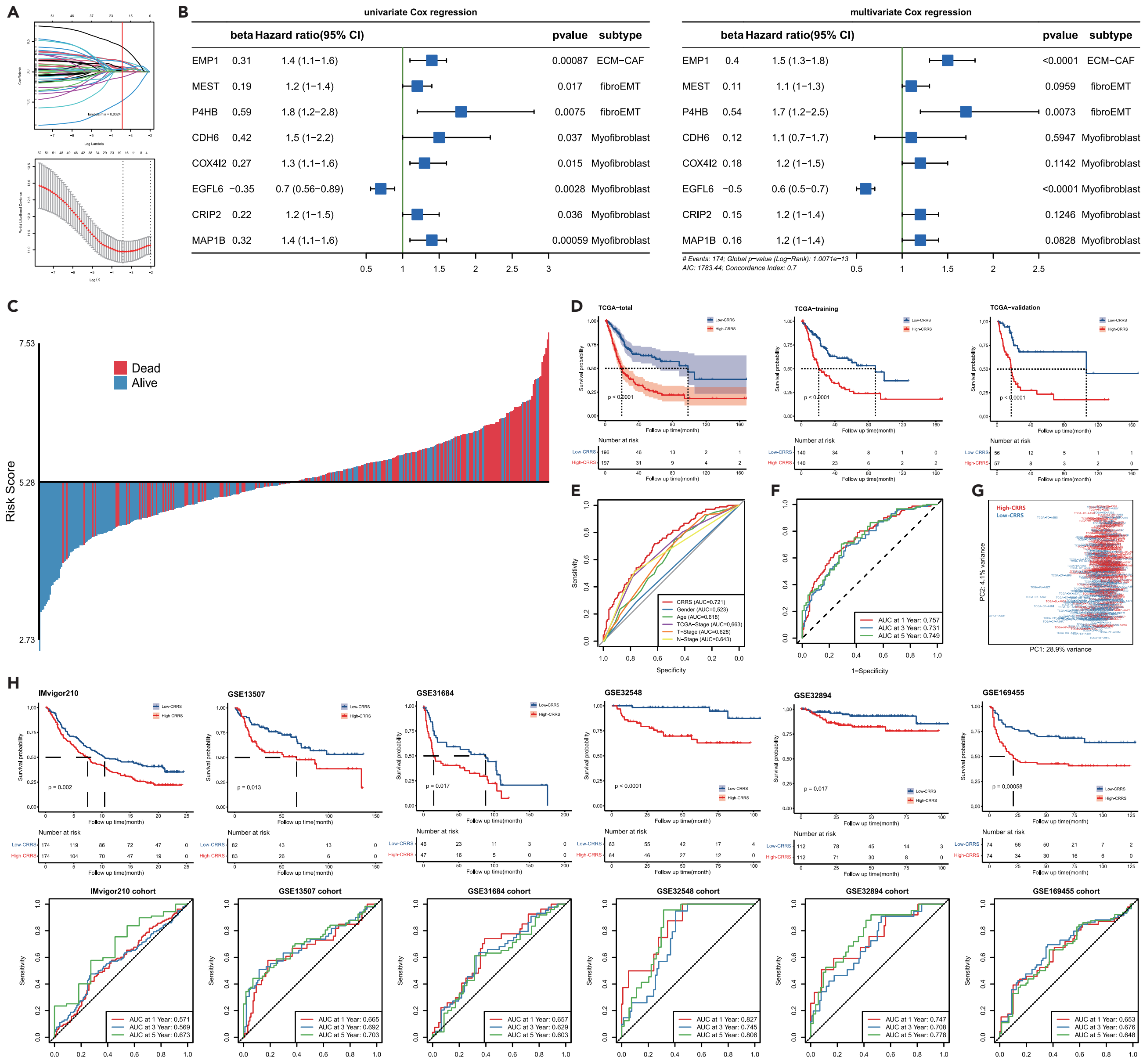

Figure 2. Identification of a CAFs-related risk score associated with prognosis of BLCA

(A) LASSO regression of prognostic CAF-related markers and partial likelihood deviance were plotted versus log (lambda).

(B) Univariate and multivariate Cox regression of eight CAF-related markers in TCGA cohort (n = 393, hazard ratio with 95% confidence interval).

(C) The survival status of high- and low-CRRS groups.

(D) The Kaplan-Meier analysis of high- and low-CRRS groups in TCGA total (n = 393), training (n = 280) and validation (n = 113) cohorts (Log rank test).

(E) ROC curves of CRRS and clinical characteristics.

(F) Time dependent ROC curve of CRRS in 1-, 3-, and 5-year survival prediction.

(G) Principal component analysis was performed on the TCGA-BLCA log(tpm+1) data in the high- and low-CRRS groups.

(H) Kaplan-Meier analysis and time dependent ROC curves of CRRS in IMvigor210 (n = 348), GSE13507 (n = 165), GSE31684 (n = 93), GSE32548 (n = 127), GSE32894 (n = 224), and GSE169455 (n = 148) cohorts (Log rank test).

See also Figures S3 and S4; Tables S1 and S4.

1-, 3-, and 5-year OS were 0.757, 0.731, and 0.749, respectively (Figure 2F). Principal component analysis (PCA) showed that patients in different CRRS groups were well classified into two clusters (Figure 2G).

We further validated the prognostic value of CRRS in several external BLCA cohorts. The survival and ROC curve obtained showed that CRRS was a valuable predictor in the IMvigor210 cohort, GSE13507 cohort, GSE31684 cohort, GSE32548 cohort, GSE32894 cohort, and

GSE169455 cohort (Figure 2H). To eliminate the influence of confounding factors, we conducted a subgroup survival analysis according to age and stage. In these subgroups, CRRS still showed excellent predictive efficacy (Figure S4).

Constructing a nomogram based on CAF-related risk score

Subsequently, we constructed a nomogram based on the multivariate Cox regression result of CRRS, age, and pathologic stage in the TCGA cohort (Figure 3A). As illustrated in Figure 3B, the nomogram showed a comparatively high accuracy for the OS of BLCA patients by the calibration curves. Univariate and multivariate Cox regression of the TCGA-BLCA cohort (Figure 3C) and six external validation cohorts (Figure 3D) demonstrated that CRRS could act as an independent prognostic factor for patients with BLCA. In addition, decision curve analysis (DCA) suggested that the nomogram remarkably outperformed a single independent predictive parameter (Figure 3E). In summary, our integrated nomogram had a potential prognostic predictive value for BLCA patients.

Delineation of molecular characteristics in BLCA

As shown in Figure 4A, the correlation between CRRS and clinicopathological characteristics was displayed. The BLCA patients with age $\geq$60 and later TNM stage had higher risk scores in the TCGA cohort (Figure 4D). Significantly, the CRRS regulatory genes were differentially expressed in high- and low-CRRS groups. Furthermore, the oncogenic pathways were significantly enriched in the high-CRRS group, including *p*-EMT, hypoxia, DNA replication and TGF-β signaling, whereas the pathways of differentiation and FGFR3-coexpressed genes were enriched in the low-CRRS group (Figure 4B). Targeting these pathways may benefit patients in the high-CRRS group. We further explored the correlation between CRRS and seven classic molecular subtyping classifications of BLCA. The BLCA patients with low-CRRS were more likely to be the luminal subtype and the high-CRRS group indicated a basal subtype (Figure 4C). We further explored the transcriptomic differences between the high- and low-CRRS groups. The regulon activities of 23 transcription factors that were related to BLCA were significantly different. The transcription factors associated with tumor promotion including FOXM1, EGFR, HIF1A, STAT3 and FGFR1 were higher in the high-CRRS group than in the low-CRRS group (Figure 4E). Moreover, regulon activity associated with chromatin remodeling suggested that the high-CRRS group exhibited higher activity of EGR1, FLI1, SIRT4, HDAC4, HDAC5, HDAC9, DNMT1 and YAP1 (Figure S5).

The correlation between EMT and CRRS in BLCA

The biological process of EMT involves epithelial cells acquiring a mesenchymal characteristic, which is an important step for enhanced myometrial invasion and metastasis capacity of BLCA cells.[13] Next, we investigated the correlation between EMT and CRRS. Notably, our results indicated that the CRRS level in muscle-invasive bladder cancer (MIBC) was higher than in non-muscle-invasive bladder cancer (NMIBC), and the proportion of MIBC in the high-CRRS group was higher than that of the low-CRRS group (Figure 5A). Consistently, the EMT score was significantly increased in the high-CRRS group (Figure 5B). The CRRS was also positively correlated to the *p*-EMT score (Figure 5C, $R = 0.45$, p value $< 2.2e-16$). Besides, the survival analysis also suggested that the patients with high-CRRS had shorter progression-free survival than the patients with low-CRRS in the UROMOL2016 cohort ($p = 0.0025$) (Figure 5D). Interestingly, none of the 106 patients with lower CRRS progressed from NMIBC to MIBC. Furthermore, we performed GSEA enrichment analysis and discovered that EMT signaling pathway was remarkably activated in the high-CRRS group (Figures 5E and 5F).

The immune landscape and immunotherapy response in different CRRS groups

The tumor immune microenvironment influenced tumor progression and the efficacy of immunotherapy. We first investigated the relevance between CRRS and the infiltration of immune cells. Our results revealed that several anticancer tumor-infiltrating immune cells (TIICs) including CD8+T cells and activated dendritic cells were significantly downregulated in the high-CRRS group (Figures 6A–6C and S6A). Moreover, in our study, CRRS was discovered to be positively correlated with TIDE score, T cell exclusion and dysfunction score in the TCGA cohort, which indicated that the high-CRRS group represented a non-inflamed phenotype (Figure 6D). These results were further validated in the external imvigor210 cohort (Figure S6B). Subsequently, samples in the IMvigor210 cohort were classified into deserted, excluded, and inflamed phenotypes, according to the spatial distribution of immune cells.[14] The patients with non-inflamed phenotype had a higher T cell exclusion score, and the TIDE score in the ICB resistance group was higher than that of the ICB response group (Figures S6C–S6D). The groups were identified as complete response (CR), partial response (PR), stable disease (SD), and progressive disease (PD). As shown in Figure 6E, high CRRS was positively associated with excluded phenotypes and predicted a worse ICB response in BLCA. The ROC curves also suggested that the CRRS had a strong predictive capacity for ICB complete response (Figure 6F). Consistently, the result of the melanoma immunotherapy cohort GSE91061 also showed that the remission rate of immunotherapy in the high-CRRS group was significantly lower than that in the low-CRRS group (Figure 6G).

Identification of DEGs in the high- and low-CRRS groups

In this study, 642 common DEGs were obtained by intersecting 3 different methods under the criteria with $|\log2FC| > 1.0$ and FDR <0.05 (Figure 7A; Table S5). Then, the most significant DEGs ($|\log2FC| > 2$ and FDR <0.001) were presented in the volcano plot (Figure 7B). The enrichment results based on KEGG and GO databases indicated that the upregulated DEGs were enriched in the biological pathways such as ECM-receptor interaction, focal adhesion, collagen-containing extracellular matrix and mesenchymal-epithelial cell signaling (Figures 7C and 7D). The GSEA analysis was consistent with the KEGG and GO enrichment results (Figure 7E). Furthermore, we conducted a protein-protein interaction (PPI) network analysis of the up-regulated DEGs using the STRING database and cytoscape software (Figure 7F).

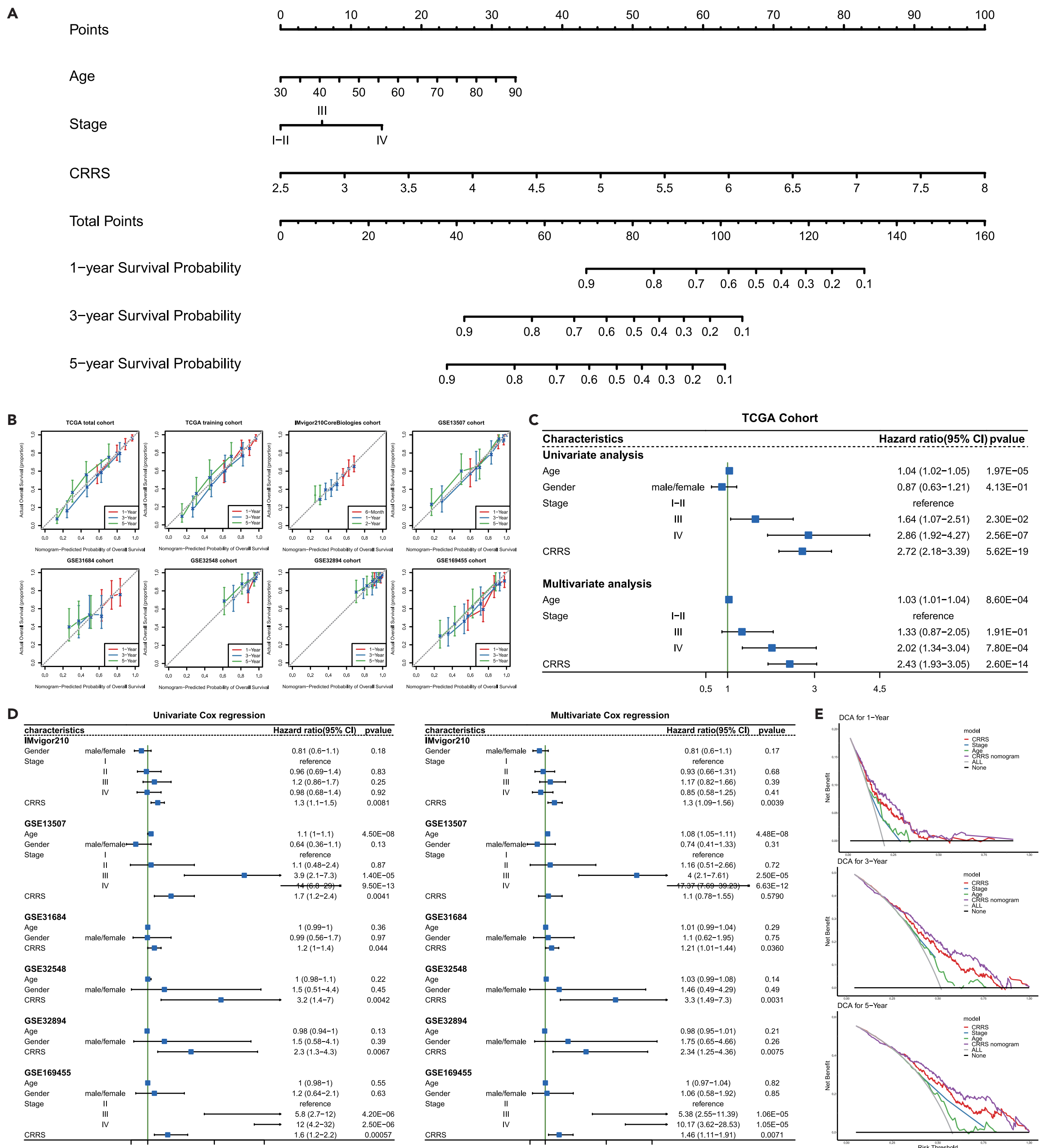

Figure 3. The nomogram to predict the prognosis of BLCA patients by integrating CRRS and clinicopathological characteristics

(A) The nomogram for predicting 1-year, 3-year, and 5-year survival probability.

(B) Calibration curves of the nomogram in TCGA-BLCA and six external validation cohorts including IMvigor210, GSE13507, GSE31684, GSE32548, GSE32894 and GSE169455. The x axis represents the nomogram - predicted probability and y axis represents the actual survival probability.

(C) Forest plot of univariate and multivariate Cox regression in TCGA-BLCA cohort (n = 393, hazard ratio with 95% confidence interval).

(D) Forest plot of univariate and multivariate Cox regression in six external validation cohorts (hazard ratio with 95% confidence interval).

(E) Decision curve analysis was performed to assess the accuracy of the nomogram.

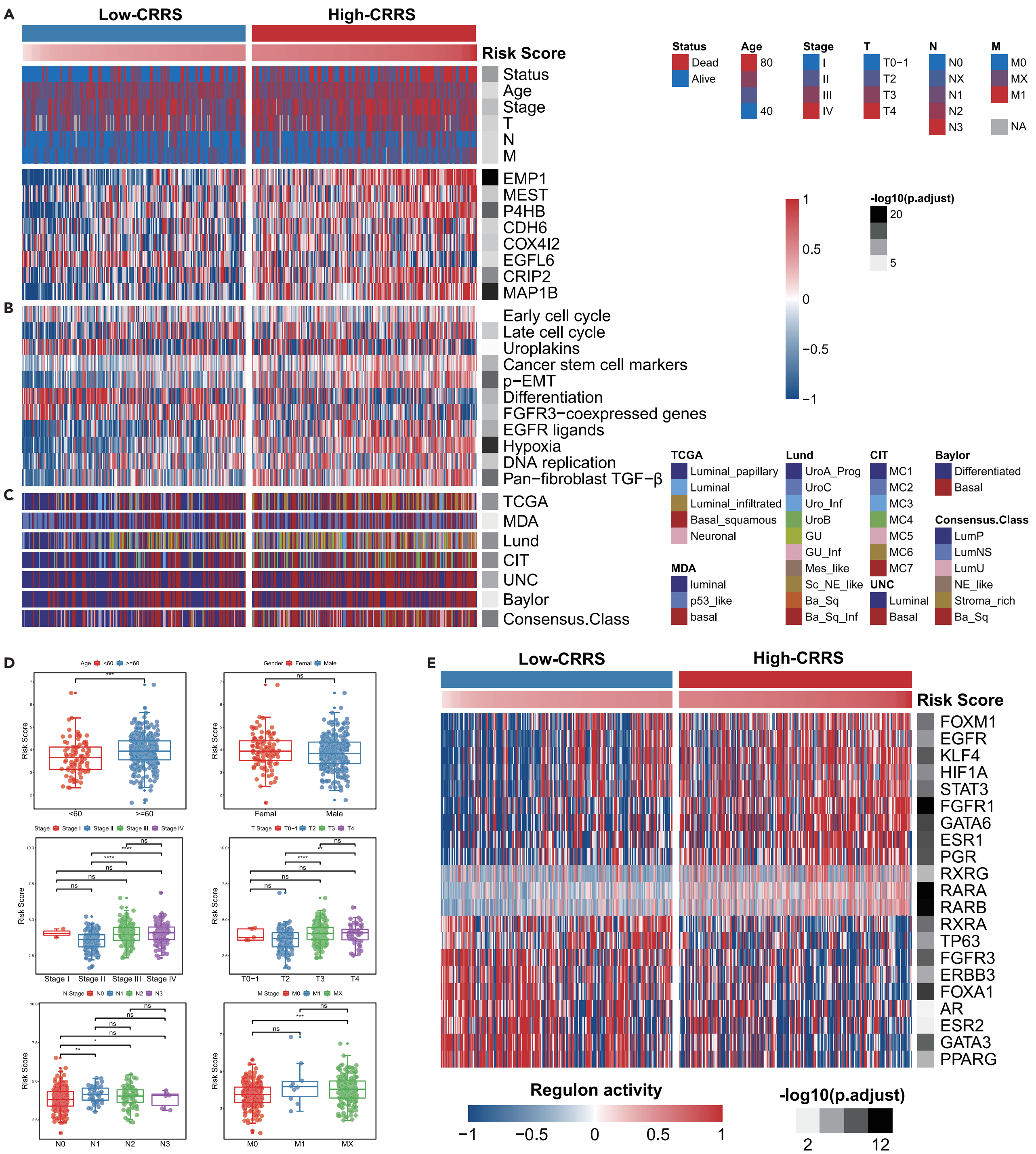

Figure 4. Distinction of clinicopathological and molecular characteristics in high- and low-CRRS groups

(A) The correlation between CRRS and clinicopathological characteristics, and the expression of 8 CAF-related markers (n = 393, Wilcoxon rank-sum test for continuous variables, Chi-square test for categorical variables).

(B) The heatmap showed differentially enriched biological pathways (n = 393, Wilcoxon rank-sum test).

(C) The association between CRRS and the molecular subtypes in seven classic systems (n = 393, Chi-square test).

(D) CRRS of TCGA-BLCA cohort with different clinicopathological features (n = 393, Wilcoxon rank-sum test and Nemenyi test).

(E) The heatmap of regulon activity profiles for 23 transcription factors in high and low-CRRS groups (n = 393, Wilcoxon rank-sum test). FDR-adjustment was performed for adjusted p values. *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001; ns: no significant. Data were represented as mean ± SEM. See also Figure S5 and Table S6.

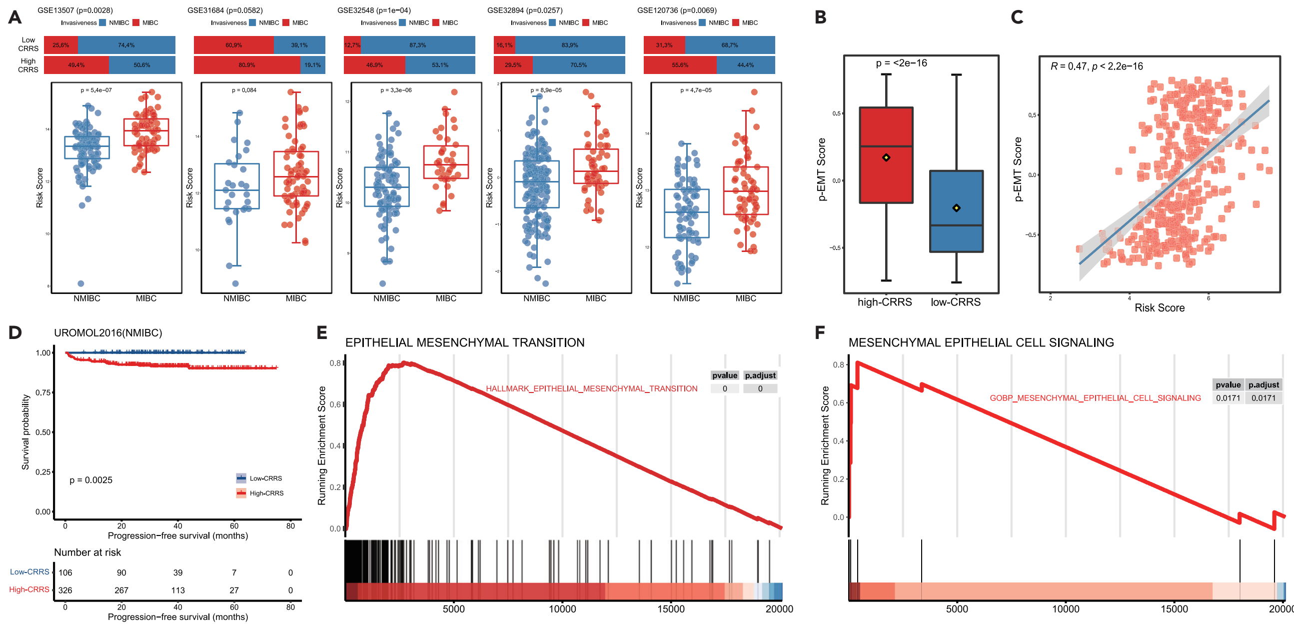

Figure 5. CRRS was positively correlated with epithelial mesenchymal transition in BLCA

(A) The CRRS in muscle-invasive bladder cancer (MIBC) and non-muscle-invasive bladder cancer (NMIBC) in GSE13507 (n = 165), GSE31684 (n = 93), GSE32548 (n = 127), GSE32894 (n = 224), and GSE120736 (n = 139) cohorts (Chi-square test for categorical variable, Wilcoxon rank-sum test for continuous variable).
(B) *p*-EMT score in high- and low-CRRS groups (n = 393, Wilcoxon rank-sum test).
(C) Scatterplot of the correlation between CRRS and *p*-EMT score (n = 393, Spearman rank correlation test).
(D) KM survival analysis in UROMOL2016 cohort (n = 432, Log rank test).
(E) GSEA plot of "HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION" based on TCGA-BLCA cohort.
(F) GSEA plot of "MESENCHYMAL EPITHELIAL CELL SIGNALING". Data were represented as mean ± SEM.
See also Table S6.

Subsequently, the top 10 differentially mutated genes between the low- and high-CRRS groups were summarized in Figure 7G. The mutation rates of RB1 and APOB were significantly increased in the high-CRRS group (Figure 7G). Besides, we detected cancer-driver genes in the high-CRRS group including GNA13, TTYH1, KRAS, CDKN2A and NFE2L2, whereas none of these genes was detected in the low-CRRS group (Figure 7H). The difference of chromosomal copy number variation sites was presented in Figure 7I. These results may explain why the high-CRRS group had a poorer prognosis and higher pathological stage.

Validating the roles of the CRRS in the Xiangya real-world cohort

In our Xiangya real-world cohort, we further validated the role of the CAFs-related risk score in predicting prognosis. As shown in Figures 8A and 8B, patients with high-CRRS had shorter overall survival than those with low-CRRS scores. Univariate and multivariate cox regression analysis also identified that CRRS was an independent prognostic factor in the Xiangya real-world cohort (Figure 8C). The AUCs of ROC curves in CRRS predicting 1-, 3-, and 4-year OS were 0.846, 0.71, and 0.788, respectively (Figure 8D). In this study, we selected EMP1, P4HB and EGFL6 for further experimental verification, due to the high weight of these three genes in the CRRS formula (0.4 for EMP1, 0.54 for P4HB and −0.5 for EGFL6). The results of immunofluorescence showed that EMP1, P4HB and EGFL6 were significant markers for specific CAF subtypes (Figure 8E). In addition, we analyzed the association between CRRS and the level of immune cell infiltration, and the expression of immune checkpoints. The high-CRRS group had lower expression of CD8, PD-L1 and CTLA4 compared with the low-CRRS group, which indicates that CRRS was significantly correlated with a non-inflamed phenotype (Figure 8F).

DISCUSSION

Globally, BLCA is the fourth most common malignant tumor in the male population.[15] It is widely known that BLCA is a highly heterogeneous tumor with a high progression rate and seriously threatens human health. The mechanism of BLCA progression and drug resistance is still unclear and worth further exploration.

TME refers to a complex niche in which the change of any component including immune cells, fibroblasts, extracellular matrix, chemokines and cytokines will affect the growth, metastasis, and therapeutic response of tumor cells. CAF is one of the main components in the TME of BLCA. A previous study suggested that iCAFs promoted the proliferation of BLCA cells and were significantly correlated with a poor OS of BLCA patients.[10] The research by Xingbo et al. indicated that CAFs could lead to cisplatin resistance in BLCA cells by regulating ERβ/Bcl-2 signaling pathway.[16] In addition, Zheng et al. discovered that CAFs-derived MFAP5 resulted in the activation of NOTCH2/HEY1 signaling

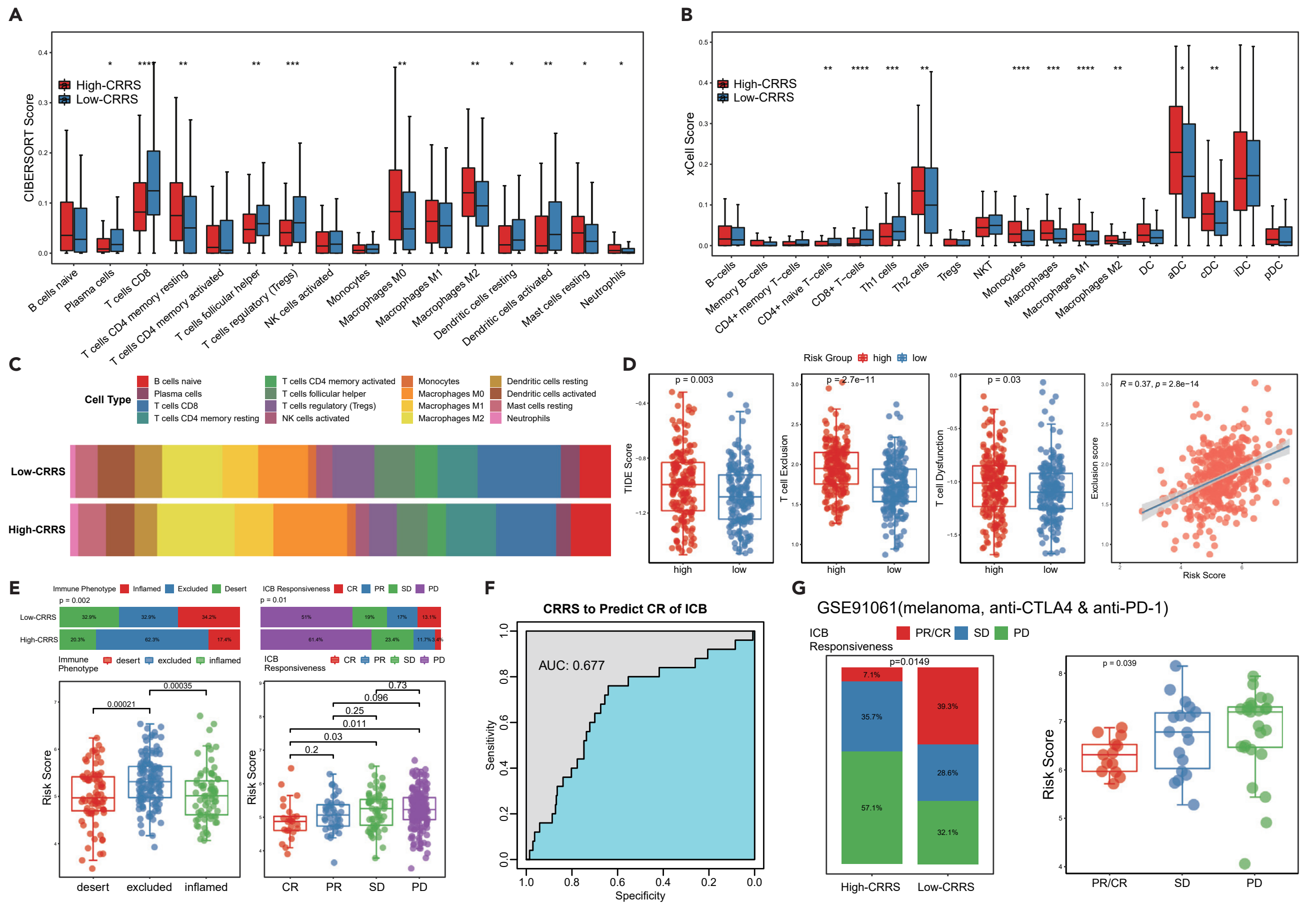

Figure 6. The role of CRRS in immune microenvironment and immunotherapy

(A) The correlation between CRRS and the infiltration levels of immune cells based on CIBERSORT algorithm (n = 393, Wilcoxon rank-sum test).

(B) The association between CRRS and the infiltration levels of immune cells based on xCell algorithm (n = 393, Wilcoxon rank-sum test).

(C) Proportion of immune cells in high- and low-CRRS groups based on CIBERSORT algorithm.

(D) The correlation between CRRS and TIDE score, T cell exclusion and T cell dysfunction score (n = 393, Wilcoxon rank-sum test and Spearman rank correlation test).

(E) The CRRS in different immune phenotype, and the correlation between CRRS and responsiveness to immunotherapy in IMvigor210 cohort (n = 348, Nemenyi test).

(F) The AUC curve evaluating the accuracy for predicting ICB complete response by CRRS.

(G) The correlation between CRRS and ICB responsiveness in GSE91061 cohort (n = 56, Chi-square test and Kruskal-Wallis test). *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$. Data were represented as mean ± SEM.

See also Figure S6 and Table S1.

pathway to promote BLCA proliferation and metastasis.[17] However, the majority of research focused on the role of a single CAF-related gene in BLCA and the comprehensive analysis of multiple CAF-related gene signatures was worth further exploration.

Several CAF-related prognostic models for BLCA have been developed.[18–20] However, all of these models were based on CAF markers obtained from the known literatures or CAF infiltration assessed by R packages common to all types of cancer. In this study, we successfully identified three distinct CAF subtypes that were not defined before and characterized 457 CAFs-related markers in BLCA by integrating reliable scRNA-seq data and bulk RNA-seq data, which could better reveal the TME heterogeneity of BLCA. Subsequently, the specific expression of these markers in corresponding CAF subtypes was verified in the validating scRNA-seq dataset of BLCA.[21] Since the differences in the source for activated fibroblasts could result in a heterogeneous population in tumors,[22] a prognostic model which consisted of numerous marker genes would be more effective compared with single biomarker. To make our model more accurate and plausible, we not only performed conventional methods such as Cox regression and LASSO regression to screen prognostic markers, but also combined a total of 102714 signatures using markers obtained from LASSO regression and evaluated each signature based on three cohorts (TCGA-BLCA, IMvigor210 and GSE13507) to select the optimal model. Therefore, we constructed a robust signature that consisted of eight CAF-related markers, which we called CAF-related risk score (CRRS), to predict the prognosis, immune cell infiltration and immunotherapy response for BLCA patients. More significantly, our CRRS performed well in survival analysis, ROC curves, univariate and multivariate Cox

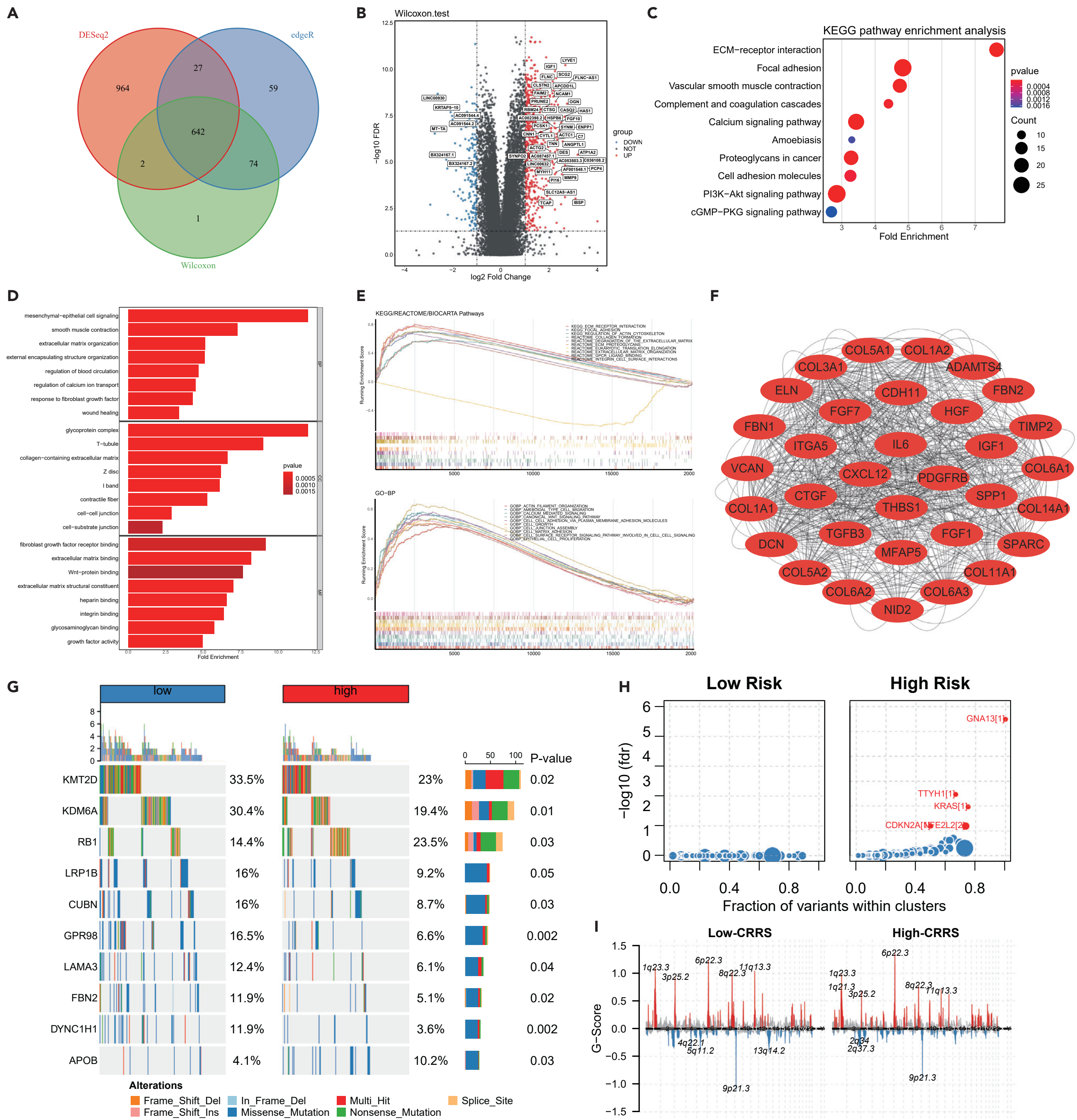

Figure 7. DEGs and genomic alterations in high- and low-CRRS groups

(A) Venn diagram of differentially expressed genes (DEGs) based on DEGSeq2, edgeR and Wilcoxon test.

(B) The volcano plot of DEGs, labels showed DEGs with the most significant expression differences (log2Foldchange >2 & FDR <0.001).

(C) Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis of upregulated DEGs.

(D) Gene Ontology (GO) analysis of upregulated DEGs.

(E) Gene Set Enrichment Analysis (GSEA) between high- and low-CRRS groups of GO, KEGG and REACTOME pathways.

(F) The protein - protein interaction (PPI) network of hubgenes of upregulated DEGs via STRING website.

(G) The differentially mutated genes in low- and high-CRRS groups respectively.

(H) Significant cancer driver genes in low- and high-CRRS groups.

(I) Chromosomal copy number variation in low- and high-CRRS groups. See also Table S5.

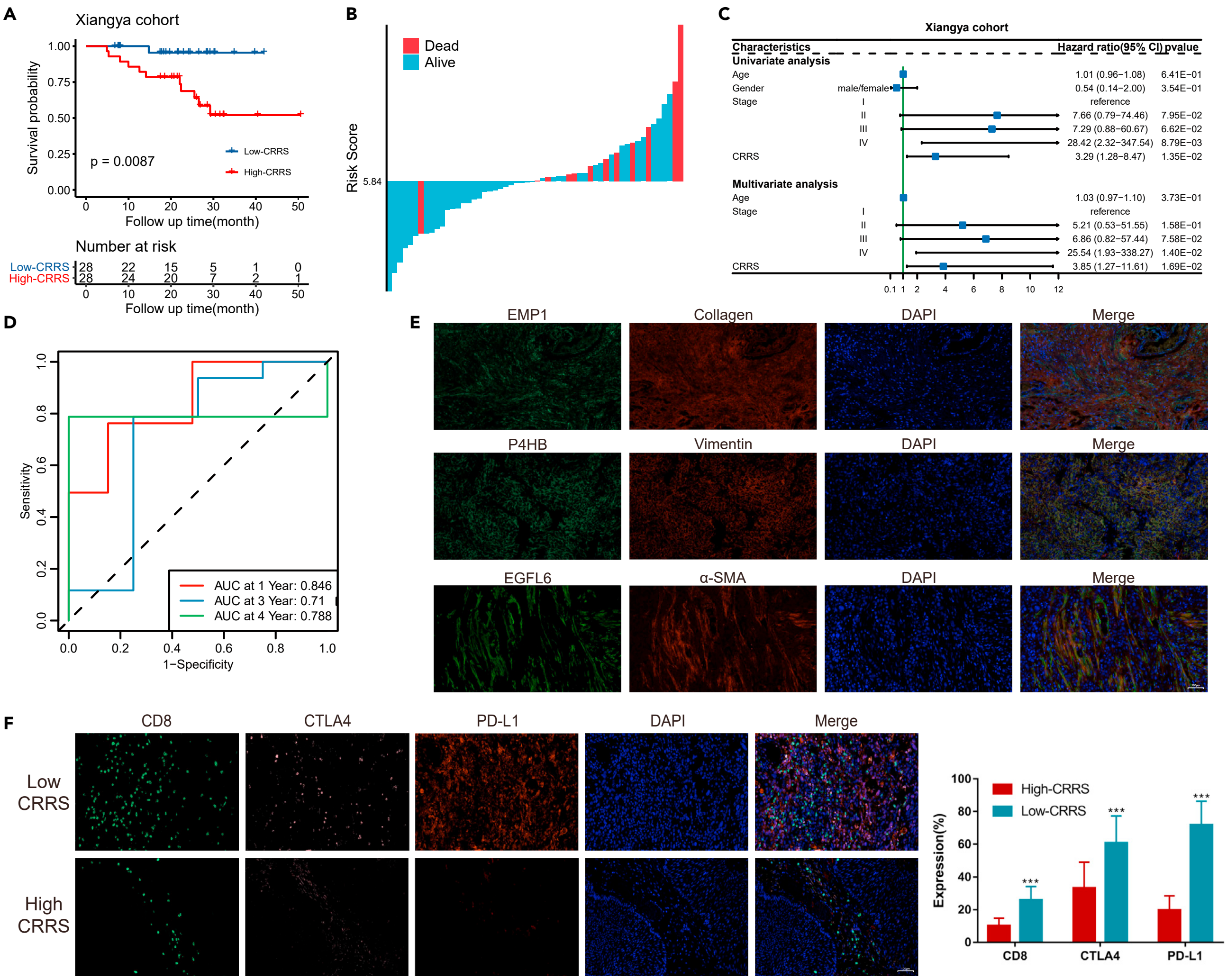

Figure 8. Roles of CAFs-related risk score in the Xiangya real-world cohort

(A) KM survival analysis of high- and low-CRRS group (n = 56, Log rank test, p = 0.0087).

(B) Histogram showing the correlation between survival status and CAFs-related risk score.

(C) Forest plot showing the independent prognostic value of CAFs-related risk score in the BLCA Xiangya real-world cohort (hazard ratio with 95% confidence interval).

(D) ROC curves showing the predictive accuracy of CAFs-related risk score for overall survival in Xiangya real-world cohort.

(E) Double-staining of EMP1/Collagen, P4HB/Vimentin, and EGFL6/α-SMA (Scale bar = 100 μm).

(F) Expression of CD8, PD-L1, and CTLA4 in high- and low-CRRS group were detected by immunofluorescence (n = 56, Student's *t* test) (Scale bar = 100 μm, data were represented as mean ± SEM). See also Table S1.

regression, nomogram, clinical decision analysis (DCA), and subgroup survival analysis in a total of 2042 patients including 9 public datasets and the Xiangya real-world cohort, whereas other models had never been validated the efficacy in such a large number of samples.

Next, we explored the 23 transcriptional regulatory factors previously reported for BLCA[23] and 78 candidate regulators associated with chromatin remodeling in cancer.[24] The analysis indicated that the regulon activity profiles for transcription and chromatin remodeling were obviously varied between the high- and low-CRRS groups. These findings confirmed the significance of transcriptional driver and chromatin remodeling events in BLCA progression, and were capable of discriminating subpopulations and choosing treatment.[25] The mutational burden, genomic structure alterations, treatment and prognosis in MIBC and NMIBC were remarkably distinct.[26] We further found that CRRS was higher in MIBC patients than that in NMIBC patients, and CRRS were significantly correlated with the EMT status of BLCA. Previous research reported that CAFs could induce the EMT characteristics of cancer cells to promote tumor growth and metastasis.[27,28] The DEGs and enrichment analyses between the high- and low-CRRS groups in the TCGA cohort also suggested that CRRS was associated with extracellular matrix secretion and epithelial-mesenchymal transformation signaling. More convincingly, none of the patients with low CRRS progressed from NMIBC to MIBC in the UROMOL2016 cohort.

In the present study, we discovered that the infiltration of $CD8^+$ T cells was lower in the high-CRRS group. The high levels of $CD8^+$ T cells in tumor tissues were associated with an effective response to immunotherapy.[29] The TIDE algorithm could be used to predict ICB response by simulating tumor immune escape according to the signatures of T cell dysfunction and T cell exclusion.[30] Daniel et al. divided tumors into immune-inflamed phenotype, immune-excluded phenotype and immune-desert phenotype based on the infiltration and distribution of immune cells, which were significantly correlated with the response rates of immunotherapy.[14] Our CAFs-related risk score indicated that a high-CRRS group with non-inflamed phenotype rarely responded to anti-PD-L1/PD-1 and anti-CTLA4 therapy. Consistently, the results of immunofluorescence in the Xiangya real-world cohort further validated the low expression levels of CD8, PD-L1 and CTLA4 in the high-CRRS group.

Limitations of the study

Although this study already performed multidimensional analyses using scRNA-seq data combined with large bulk transcriptomics data, there were still several limitations in this study. First, due to intratumor genetic heterogeneity and the sample bias, more single-cell data from different BLCA patients are required to validate our findings. In addition, the CAF subtypes and CAF-related risk signatures were acquired from retrospective data based on public datasets. Consequently, the CRRS should be validated in more prospective cohorts in the future. Finally, additional researches should be performed to investigate the molecular mechanisms of these CAF-related gene signatures in the development of BLCA.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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AUTHORS CONTRIBUTIONS

H.S. and Y.C. designed and supervised the study. Y.Q. and Y.C. did the bioinformatics analysis and experiment. Y.L., Y.H., J.T., C.C., and E.S. performed the literature search and collected the data. P.L., G.D., Z.F., Y.P., Y.L., J.M., and S.Z. analyzed the data. Y.Q. and Y.C. wrote and revised the manuscript. All authors contributed to the manuscript.

DECLARATION OF INTERESTS

The authors declare that they have no conflicts of interest with the contents of this article.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-EMP1	Abmart	Cat# PU946505
Rabbit polyclonal anti-collagen	ZEN BIO	Cat# 501352
Rabbit monoclonal anti-P4HB	ZEN BIO	Cat# R25241
Mouse monoclonal anti-vimentin	Abcam	Cat# ab8069; RRID: AB_306239
Rabbit polyclonal anti- EGFL6	Abmart	Cat# PH1602
Rabbit monoclonal anti-α-SMA	HUABIO	Cat# ET1607-53
Rabbit monoclonal anti-CD8	Cell Signaling Technology	Cat# 85336; RRID: AB_2800052
Rabbit monoclonal anti-CTLA4	Abcam	Cat# ab237712; RRID: AB_2905652
Rabbit monoclonal anti-PD-L1	Abcam	Cat# ab205921; RRID: AB_2687878
Biological samples		
Human bladder cancer specimens	Department of urology and pathology in Xiangya Hospital	N/A
Critical commercial assays		
3-Color Multiple fluorescence Kit	AiFang biological	Cat# AFIHC024
Deposited data		
Single cell RNA-seq data for training	Lai et al.[11]	GSE135337
Single cell RNA-seq data for validation	Li et al.[21]	GSE130001
TCGA-BLCA cohort	TCGA data portal	https://portal.gdc.cancer.gov/repository
GEO validation cohort 1 for bladder cancer	Kim et al.[31]	GSE13507
GEO validation cohort 2 for bladder cancer	Riester et al.[32]	GSE31684
GEO validation cohort 3 for bladder cancer	Lindgren et al.[33]	GSE32548
GEO validation cohort 4 for bladder cancer	Sjödahl et al.[34]	GSE32894
GEO validation cohort 5 for bladder cancer	Sjödahl et al.[35]	GSE169455
GEO validation cohort 6 for bladder cancer	NCBI GEO	GSE120736
Anti-PD-L1 immunotherapy cohort for bladder cancer (IMvigor210CoreBiologies)	Mariathasan et al.[36]	http://research-pub.gene.com/IMvigor210CoreBiologies/
UROMOL2016 cohort for NMIBC	Hedegaard et al.; Lindskrog et al.[37,38]	https://www.nature.com/articles/s41467-021-22465-w#Sec38
Xiangya real world cohort for bladder cancer	Hu et al.[39]	GSE188715
Immunotherapy cohort for melanoma	Riaz et al.[40]	GSE91061
Software and algorithms		
R version 4.1.3	https://www.r-project.org/	https://www.r-project.org/
Seurat v4.1.1	Butler et al.[41]	http://satijalab.org/seurat/
SingleR v1.8.1	Aran et al.[42]	https://github.com/dviraran/SingleR
monocle v2.22.0	Qiu et al.[43]	http://cole-trapnell-lab.github.io/monocle-release/
CellChat v1.4.0	Jin et al.[44]	http://www.cellchat.org/
survival v3.3-1	CRAN	https://search.r-project.org/
glmnet v4.1-4	CRAN	https://search.r-project.org/
rms v6.3-0	CRAN	https://search.r-project.org/
ggDCA v1.2	Github	https://github.com/yikeshu0611/ggDCA
BLCAsubtyping v2.1.1	Kamoun et al.[45]	https://github.com/cit-bioinfo/BLCAsubtyping

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
consensusMIBC v1.1.0	Kamoun et al.[45]	https://github.com/cit-bioinfo/consensusMIBC
clusterProfiler v4.2.2	Wu et al.[46]	https://github.com/YuLab-SMU/clusterProfiler
CIBERSORT	Newman et al.[47]	https://rdrr.io/github/singha53/amritr/src/R/supportFunc_cibersort.R
xCell v1.1.0	Aran et al.[48]	https://github.com/dviraran/xCell
DESeq2 v1.34.0	Love et al.[49]	http://www.bioconductor.org/packages/release/bioc/html/DESeq2.html
edgeR v3.36.0	Robinson et al.[50]	http://bioconductor.org/packages/release/bioc/html/edgeR.html
RTN v2.18.0	Castro et al.[25]	http://bioconductor.org/packages/release/bioc/html/RTN.html
maftools v2.10.05	Mayakonda et al.[51]	https://github.com/PoisonAlien/maftools
GISTIC 2.0 v6.15.28	GenePattern	https://cloud.genepattern.org
Cytoscape v3.9.0	Shannon et al.[52]	https://cytoscape.org/
Code archive	This paper	N/A

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Hong Shen (hongshen2000@csu.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Single cell RNA-seq data of bladder cancer (GSE135337 and GSE130001) and clinical information and gene expression data of cohorts for GSE13507, GSE31684, GSE32548, GSE32894, GSE169455, GSE91061 and GSE120736 are available in GEO database (https://www.ncbi.nlm.nih.gov/geo/). The transcriptomic and clinical data of TCGA-BLCA cohort was retrieved from TCGA data portal (https://portal.gdc.cancer.gov/repository). UROMOL2016 cohort for NMIBC is provides in the supplementary materials of Lindskrog's study (https://www.nature.com/articles/s41467-021-22465-w#Sec38). IMvigor210CoreBiologies, an anti-PD-L1 immunotherapy cohort for bladder cancer, is deposited in R package "IMvigor210CoreBiologies". Xiangya real-world cohort for bladder cancer has been uploaded to GEO database with accession number GSE188715.
- This paper dose not report original code.
- Any additional information required to reanalyze the data reported in this study is available from the lead contact upon reasonable request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Public and Xiangya real-world cohorts

A total of 2003 patients from 9 available public datasets were collected for this study. The Cancer Genome Atlas (TCGA) cohort for bladder cancer (BLCA) was downloaded from TCGA database (https://portal.gdc.cancer.gov/repository), including RNA-seq data and the corresponding clinical information of 405 patients. After removing censored patients with follow-up times of less than 30 days, 393 patients were pseudorandomly assigned to the training dataset (n = 280) and internal validation dataset (n = 113). Five external validation cohorts with clinical information were obtained from Gene Expression Omnibus (GEO; https://www.ncbi.nlm.nih.gov/geo/), including GSE13507, GSE31684, GSE32548, GSE32894 and GSE169455 cohort. An anti-PD-L1 immunotherapy cohort of BLCA with 348 patients was retrieved from an R package "IMvigor210CoreBiologies" (http://research-pub.gene.com/IMvigor210CoreBiologies/).[36] We excluded patients with no survival information or loss to follow-up in 30 days. The immunotherapy cohort of melanoma was downloaded from GSE91061. In addition, 145 patients with information on BLCA invasiveness were retrieved from GSE120736. UROMOL2016 cohort comprised of 432 patients with non-muscle invasive bladder cancer (NMIBC) was obtained from the supplementary materials of Lindskrog's work.[37,38]

Sixty fresh bladder cancer tissues were collected from Xiangya hospital and immediately frozen using liquid nitrogen. Total RNA was extracted from the fresh tissues using TRIzol (Invitrogen, Carlsbad, CA, USA). Subsequently, NanoDrop and Agilent 2100 bioanalyzer (Thermo Fisher Scientific, MA, USA) were used to quantify the total RNA. After the construction of mRNA library, the total RNA was purified and fragmented into small pieces. Finally, we synthesized the first- and second-strand cDNA. Furthermore, the cDNA was amplified using PCR and the final library was constructed (single-stranded circular DNA). Ultimately, 56 BLCA samples were sequenced on BGISEQ-500 platform (BGI-Shenzhen, China), which we called the Xiangya real-world cohort (GEO accession, GSE188715). Clinical data and related pathological features were obtained from the patients' medical records. This study was reviewed and approved by the Medical Ethics Committee of Xiangya Hospital of Central South University.

The detailed information of these cohorts is summarized in Table S1.

Single cell RNA-seq data

Two scRNA-seq datasets for bladder cancer were obtained from GEO database with accession numbers GSE135337 and GSE130001, as the single cell discovery dataset and validation dataset, respectively.

METHOD DETAILS

Single cell RNA-seq data analysis

To get the cell marker list of cancer-associated fibroblasts (CAF) subtypes, the R package "Seurat" (v4.1.1)[53] was used to analysis scRNA-seq data according to the following steps: (1) Quality control: after removing low-quality cells with >10% mitochondrial gene expression (indicating apoptosis), <200 genes expressed or >6000 genes expressed, a total of 42440 cells were retained as high-quality cells. (2) Clustering and dimensional reduction: the expression matrix of the top 5000 genes was scaled to avoid the dominance of highly expressed genes. To cover the highest variance of the dataset, 20 principal components (PCs) were set based on an elbow plot of the first 40 PCs. Then, 42440 cells were divided into 16 clusters using functions "FindNeighbors" and "FindClusters" (resolution = 0.5). For visualization, uniform manifold approximation and projection (UMAP) was performed. (3) Cell type annotation: based on the combination of the expressions of cell type-specific markers given in the original reports[11] and the R package "SingleR" (v1.8.1), clusters were assigned to five major cell types, including epithelium, fibroblast, macrophage, T cell and endothelium. Then, we identified significant marker genes for different cell types. (4) CAF subtypes identification: interestingly, we found that cluster 13 was distributed in both epithelial and fibroblast clusters on the UMAP dimplot. The epithelial part of cluster 13 highly expressed epithelial markers, while the fibroblastic part of cluster 13 expressed mesenchymal markers with a decreased but not absent expression of epithelial markers. Subsequently, we calculated the EMT Score and Epithelial Score of the two parts of cluster 13 based on the EMT and Epithelial gene set acquired from MSigDB (https://www.gsea-msigdb.org/gsea/msigdb/index.jsp). Significantly, cluster 13 revealed the process of EMT. Thus, we resolved cluster 13 into two parts, epithelial part (named epiEMT) and fibroblastic part (named fibroEMT). Based on the makers of CAF subtypes previously reported[54,55] and the fibroEMT part we isolated, 10 clusters were identified and assigned to three CAF subclusters, ECM-CAF, myofibroblast and fibroEMT. (5) Acquisition of markers of CAF subtypes: according to the criteria of log2FC > 1 and adjusted p value <0.05, 457 markers of three CAF subtypes (169 for ECM-CAF, 103 for fibroEMT, and 185 for myofibroblast) were calculated for further study (see Table S2). (6) Trajectory analysis: we performed cell trajectory analysis for epiEMT and CAF subtypes via R package "monocle" (v2.22.0) based on the standard workflow.[43] Briefly, monocle subject was constructed using "newCellDataSet" function. Differentially expressed genes between CAF subtypes were selected for subsequent pseudotime analysis. Defining the cells of epiEMT as "root cell", dimension reduction and identification of orders of cells were performed via "reduceDimension" and "orderCells" function. (7) Cell-cell communication network construction: to investigate intercellular crosstalk between CAF subtypes and immune cells (T cells and macrophage), we applied the R package "CellChat" (v1.4.0). Commonly, the expression matrix of CAF subtypes and immune cells was used to create CellChat subject. The ligand-receptor pairs of "ECM-Receptor" and "Cell-Cell Contact" were selected from CellChat database. Cellular communication network was inferred via the "computeCommunProb", "computeCommunProbPathway" and "aggregateNet" functions. Finally, to visualize the crosstalk between CAF subtypes and immune cells, the "netVisual_circle", "netVisual_heatmap" and "netVisual_bubble" functions were used.

CAF related risk score (CRRS) development

Based on the TCGA training cohort, the univariate Cox regression analysis of 457 markers of CAF subtypes was performed to determine the prognostic significance via R package "survival" (v3.3-1). According to the criteria of Wald.test p value <0.05, 55 markers with prognostic significance were reserved. Subsequently, these markers were included in the Least Absolute Shrinkage and Selection Operator (LASSO) regression analysis to reduce overfitting and multicollinearity via the R package "glmnet" (v4.1-4). By choosing lambda.min (0.03243242) to filter these 55 markers, 18 markers were retained. To construct prognostic signatures, we randomly arranged and combined these 18 genes and a total of 102714 signatures were obtained. To find the optimal signature, we first performed multivariate Cox regression analysis to get the global assessment indices in TCGA training cohort for each of the 102714 signatures, including p.value.log, concordance index and Akaike information criterion (AIC) value. Afterward, we calculated the risk score of patients in TCGA total cohort and two external validation cohorts (IMvigor210 and GSE13507) for each signature. For the TCGA total cohort, we calculated the area under curve (AUC) value of the receiver operating characteristic (ROC) curve at 1, 3, and 5 years; for the two external validation cohorts, we performed risk grouping to calculate the log rank p values of Kaplan-Meier survival analysis between high and low risk score. In summary, we obtained 8 synthetical assessing indices for each of the 102714 signatures (see Table S4C). After synthetically screening all signatures based on the above indices, we selected the optimal signature including 8 CAF-related marker genes: EMP1, MEST, P4HB, CDH6, COX4I2, EGFL6, CRIP2, MAP1B.

The risk score of each patient was calculated using the following equation:

$$CRRS = \sum(\beta_i \times \mathrm{Exp}_i)$$

where βi represents the multivariate Cox regression coefficient of each gene in signature, Expi represents the normalized expression value of each gene in signature (log (TPM +1) for RNA-seq data, RMA-normalized value for microarray data).

Prognostic model construction and evaluation

The Kaplan-Meier survival analysis, time-dependent ROC curve and univariate Cox regression analysis of each validation cohort were performed to evaluate the prognostic value of CRRS. In addition, a nomogram to predict cancer survival rate was plotted using the R packages "rms" (v6.3-0) based on the multivariate Cox regression analysis CRRS, together with two other clinical characteristics related to the patient's survival, age and tumor stage. The calibration curves of TCGA cohort and external validation cohorts were plotted. Decision curve analysis (DCA) of CRRS, age, tumor stage and the nomogram was implemented via the R package "ggDCA" (v1.2).

Classic molecular subtypes of bladder cancer prediction

R packages BLCAsubtyping (v2.1.1) and consensusMIBC (v1.1.0) were used to predict several molecular classification systems of TCGA-BLCA cohort, including TCGA, MDA, Lund, CIT, UNC, Baylor, and Consensus classes[23,34,.45]

Gene set enrichment analyses

We collected the gene lists of cell cycle, uroplakins, cancer stem cell, differentiation, partial epithelial-mesenchymal transition (p-EMT), hypoxia, DNA replication, EGFR ligands, EGFR3 coexpressed genes and pan-fibroblast TGF-β from previous reports[34,56] (see Table S6). Single sample gene set enrichment analysis (ssGSEA) of the gene sets above was conducted via R package "GSVA" (v1.42.0). Gene set enrichment analysis (GSEA) and over-representation analysis (ORA) of Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways were conducted via the R packages "clusterProfiler" (v4.2.2) and "org.Hs.e.g.,.db" (v3.14.0). The gene sets of GO and KEGG pathways were downloaded from MSigDB (https://www.gsea-msigdb.org/gsea/msigdb/index.jsp).

Immune infiltration evaluation and immunotherapy responsiveness prediction

To evaluate the immune cell infiltration microenvironment more reliably, CIBERSORT (based on deconvolution)[47] and xCell (based on ssGSEA)[48] were performed on the TCGA cohort. CIBERSORT was employed via the CIBERSORT.R script getting from Github (https://rdrr.io/github/singha53/amritr/src/R/supportFunc_cibersort.R), and xCell was employed via the R package "xCell" (v1.1.0). For the prediction of the clinical response of immune checkpoint blockade (ICB) therapy, we analyzed the score of T cell dysfunction and exclusion (TIDE) on the TIDE website (http://tide.dfci.harvard.edu/).[30]

Differentially expressed genes identification and protein-protein interaction analysis

When applying to the large size of human RNA-seq samples, the conventional softwares filtering differentially expressed genes (DEGs), such as DESeq2 and edgeR, have unexpectedly high false positive rates. Instead, Wilcoxon rank-sum test is recommended.[57] Therefore, in addition to using R packages DESeq2 (v1.34.0) and edgeR (v3.36.0), we also performed Wilcoxon rank-sum test and false discovery rate (FDR) to search DEGs. Under the criteria with $|\log2FC| > 1.0$ and adjusted p value <0.05, we obtained 1635 DEGs for DESeq2, 802 DEGs for edgeR, and 719 DEGs for Wilcoxon test. Afterward, we selected the intersection of the three methods and obtained 642 DEGs eventually for the downstream analysis (Table S5). We performed protein-protein interaction (PPI) on the STRING website (https://cn.string-db.org) and visualized the PPI network via the software Cytoscape (v3.9.0).

Somatic mutations and copy number alternations analysis

Somatic mutations and masked copy number segment data of TCGA-BLCA were downloaded from TCGA database (https://portal.gdc.cancer.gov/repository). The copy number variation events identification was performed using GISTIC 2.0 (v6.15.28) (https://cloud.genepattern.org). Different somatic mutation types were analyzed using the R package "maftools" (v2.10.05). To identify cancer driver genes, the oncodriveCLUST algorithm was used.[58]

Regulon activity analysis

23 transcription factors associated with bladder cancer and 78 regulators associated with chromatin remodeling of cancer reported previously[23,24] (see Table S6) were collected and transcriptional regulatory networks (regulons) were constructed using the R package "RTN" (v2.18.0).

Multiplex immunofluorescence staining

Fifty-six bladder cancer specimens corresponding to Xiangya real-world cohort sequencing data (GSE188715) were acquired from the pathology department of Xiangya Hospital. 4 μm Paraffin-embedded sections were deparaffinized. After antigen retrieval, the sections were blocked with 3% H_2O_2 and 3% BSA. Multiplexed immunofluorescence staining of EMP1 (Rabbit, 1:400, abmart, China) and the ECM - CAF marker collagen (Rabbit, 1:200, ZEN BIO, China), P4HB (Rabbit, 1:200, ZEN BIO, China) and the fibroEMT marker vimentin (Mouse, 1:200, abcam, UK), EGFL6 (Rabbit, 1:100, abmart, China) and myofibroblast marker α-SMA (Rabbit, 1:200, HUABIO, China) were performed according to the manufacturer's instructions (AiFang biological, 3-Color Multiple fluorescence Kit, China). Multiplexed immunofluorescence staining of CD8 (Rabbit, 1:400, CST, USA), CTLA4 (Rabbit, 1:500, abcam, UK), and PD-L1 (Rabbit, 1:100, abcam, UK) were performed according to the manufacturer's instructions (AiFang biological, 3-Color Multiple fluorescence Kit, China). The images were captured using a Pannoramic MIDI microscope.

QUANTIFICATION AND STATISTICAL ANALYSIS

Wilcoxon rank-sum test was applied to compare continuous variables and Fisher's exact test and chi-square test were applied for categorical variables. Univariate and multivariate Cox Proportional-Hazards analyses and LASSO regression were performed to construct the prognostic signature using the R package "survival" and "glmnet". Kaplan-Meier survival analysis and log rank test were used to compare survival curves. ROC and time-dependent ROC curves were performed using the R packages "pROC" and "timeROC". Significance was considered with p value <0.05, and FDR-adjustment of p values was performed to control for multiple testing. All statistical analyses were conducted with R software (v4.1.3).